

Maximization of a Linear Function of Variables Subject to Linear Inequalities

George B. Dantzig

September 17, 2026

The general problem indicated in the title is easily transformed, by any one of several methods, to one which maximizes a linear form of non-negative variables subject to a system of linear equalities. For example, consider the linear inequality $ax + by + c < 0$. The linear inequality can be replaced by a linear equality in nonnegative variables by writing, instead $a(x_1 - x_2) + b(y_1 - y_2) + c - z = 0$, where $x_1 \geq 0$, $x_2 \geq 0$, $y_1 \geq 0$, $y_2 \geq 0$, $z \geq 0$. The basic problem throughout this chapter will be consider in the following form:

Problem: Find the vlaues of $\lambda_1, \lambda_2, \dots, \lambda_n$ which maximize the linear form

$$(1) \quad \lambda_1 c_1 + \lambda_2 c_2 + \dots + \lambda_n c_n$$

subject to the conditions that

$$(2) \quad \lambda_j \geq 0 \quad (j = 1, 2, \dots, n)$$